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Augmented Recursive Sliding Mode Observer Based Adaptive Terminal Sliding Mode Controller for PMSM Drives

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Abstract

Time-varying lumped disturbance and measurement noise are primary obstacles that restrict the control performance of permanent magnet synchronous motor (PMSM) drives. To tackle these obstacles, an adaptive nonsingular terminal sliding mode (ANTSM) algorithm is combined with augmented recursive sliding mode observer (ARSMO) for PMSM speed regulation system in this paper. Generally, conventional nonsingular terminal sliding mode (NTSM) controller adopts a fixed and conservative control gain to suppress the time-varying disturbance, which will lead to unsatisfactory steady-state performance. Without requiring any information of the time-varying disturbance in advance, a novel barrier function adaptive algorithm is utilized to adjust the gain of NTSM controller online according to the amplitude of disturbance. In addition, the ARSMO is employed to estimate the total disturbance and motor speed simultaneously, thereby alleviating the negative impact of measurement noise and excessive control gain. Comprehensive experimental results verify that the proposed enhanced ANTSM strategy can optimize the dynamic performance of PMSM system without sacrificing its steady-state performance.

Keywords: permanent magnet synchronous motor; barrier function; adaptive terminal sliding mode; high-order fast finite-time extended state observer; measurement noise suppression



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1. Introduction

With the characteristics of rapid response, high tracking precision, and simple structure, permanent magnet synchronous motor (PMSM) has extensively utilized in various industry applications [1–3]. However, the control performance of PMSM system may be inevitably influenced by time-varying disturbance and measurement noise. Generally, the total disturbance generated by load torque and parameter perturbation will cause aperiodic speed fluctuation, thereby affecting the anti-disturbance property of PMSM system [4]. The measurement noise generated by position sensors may degrade the steady-state performance of PMSM drive. Hence, the research on disturbance and measurement noise suppression has drawn great attention recently [5–9].

To improve the time-varying disturbance suppression property, many advanced control algorithms have been adopted to optimize the performance of PMSM drive system, such as backstepping control [10], fuzzy control [11], active disturbance rejection control (ADRC) [12], and sliding mode control (SMC) [13–15], etc. Owing to insensitivity to parameter mismatch and strong robustness against load torque, the SMC stands out from

the aforementioned algorithms. Distinguished from the linear sliding surface used in conventional SMC, a terminal sliding mode (TSM) controller was proposed in [16] to obtain rapid response and strong anti-disturbance property. With the benefit of nonlinear sliding surface, the speed tracking error can converge to origin in finite time.

Generally, a sufficiently large control gain should be chosen to suppress the time-varying and unknown disturbance [17]. Consequently, a fixed and conservative control gain may cause undesired chattering phenomenon, thus damaging the system hardware. On the contrary, the small control gain may lead to the instability of PMSM system in some extreme cases, where the stable condition of SMC cannot be satisfied. In [18], a neural adaptive TSM control method was constructed to automatically tune the control gain of conventional TSM. However, the utilization of neural network technique heavily depends on the computing power of hardware devices, which limits its large-scale industrial applications. A fuzzy logic controller was employed in [19] to avoid serious chattering phenomenon generated by overestimated gain of TSM control algorithm. However, the formulation of fuzzy rules is based on the experience of previous parameter tuning, which brings great challenges to the reasonable selection of the control gain. With the development of control theory, a typical gain-increasing adaptive algorithm was proposed in [20] to adjust the control gain as the disturbance increases. The aim of the above adaptive algorithm is to increase the control gain until the closed-loop system stabilizes, and then the control gain is fixed. Unfortunately, this kind of adaptive law lacks the ability to reduce the controller gain as the disturbance decreases. To address this tough issue, a novel adaptive algorithm was constructed in [21] to online adjust the control gain with the change of disturbance. As the disturbance decreases, the gain will also decrease correspondingly. However, the size of convergence neighborhood depends on some priori knowledge about the upper bound of disturbance, which is unreasonable in the practical engineering system.

Even though several SMC strategies have shown significant success in suppressing time-varying disturbance, the problems of measurement noise and excessive control gain still restrict the control accuracy of PMSM drive system. In [22], the widely used disturbance observer (DOB) was adopted to estimate and compensate the total disturbance, which alleviates the control gain of NTSM controller. Nevertheless, two significant drawbacks exist in the aforementioned control strategy. On one hand, conventional DOB can only guarantee the asymptotic convergence of the estimation error, which results in low estimation accuracy. On the other hand, conventional DOB is unable to estimate the system states, which result in the inability to suppress the measurement noise. In [23], the high-order phase-locking loop observer (PLLO) was employed to estimate the actual motor speed and lumped disturbance simultaneously, which can suppress the measurement noise and improve the low-frequency disturbance rejection performance. Unfortunately, introducing the PLLO may deteriorate the stability of PMSM drive system. In [24], two cascaded extended stated observer was utilized to filter out the measurement noise without sacrificing the robustness of the closed-loop system. However, complex control structure may increase the computational burden and parameter tuning workload.

This article presents a barrier function based adaptive nonsingular terminal sliding mode (ANTSM) algorithm combined with augmented recursive sliding mode observer (ARSMO) for PMSM speed regulation system. Firstly, utilizing the barrier function to adjust the control gain of NTSM in real time, the severe chattering generated by conservative control gain can be greatly mitigated. Secondly, considering the adverse effect of measurement noise generated by position sensors, a concise and effective ARSMO is adopted to estimate the motor speed signal. Finally, the superior advantages of the proposed PMSM speed loop control strategy are demonstrated by experimental results. The major contributions of this paper mainly include:

- (1) A barrier function is exploited to adjust the control gain of the conventional NTSM controller, which is simpler and more reasonable than existing adaptive algorithms that can only ensure that the convergence region of the sliding variable is related to the upper bound of the lumped disturbance. Without requiring any priori information of time-varying disturbance, the proposed ANTSM controller can drive the speed error and maintain it in a predefined neighborhood of origin.
- (2) By proving that the closed-loop system will not diverge to infinity before the observation error of the proposed ARSMO converges to zero, the stability of the entire closed-loop system is theoretically guaranteed. Meanwhile, according to the bode diagrams obtained by the frequency-sweep method, it can be seen from the frequency domain perspective that the proposed ARSMO has satisfactory measurement noise suppression capability.

2. Mathematical Model and Conventional Control Method

2.1. PMSM Mechanical Dynamics with Disturbance

Ignoring the effects of hysteresis loss, core saturation, and eddy current, the motion equation of the PMSM system can be described by

$$\dot{\Omega}_m = -\frac{B}{J}\Omega_m + \frac{T_e}{J} - \frac{T_L}{J} \quad (1)$$

where Ω_m is the rotor angular velocity, B is the viscous frictional coefficient, T_e is the electromagnetic torque, J is the moment of inertia, and T_L denotes the load torque.

Taking the reference electromagnetic torque T_e^* as the control input, Equation (1) can be rewritten as

$$\dot{\Omega}_m = bT_e^* - b(T_L + B\Omega_m) - b(T_e^* - T_e) \quad (2)$$

where $b = \frac{1}{J}$.

Considering the parameter mismatch issue, the system model can be reconstructed as

$$\begin{aligned} \dot{\Omega}_m &= b_0T_e^* + (b - b_0)T_e^* \\ &\quad - b(T_L + B\Omega_m) - b(T_e^* - T_e) \\ &= b_0T_e^* + d_t \end{aligned} \quad (3)$$

where $b_0 = \frac{1}{J_0}$, J_0 denotes the nominal inertia, and $d_t = (b - b_0)T_e^* - b(T_L + B\Omega_m) - b(T_e^* - T_e)$ is the total disturbance.

The actual motor speed signal is typically calculated from the derivative of the rotor position, and the procedure will result in the introduction of measurement noise. Therefore, the measured motor speed signal Ω_n contaminated by high-frequency noise n_0 is shown as

$$\Omega_n = \Omega_m + n_0 \quad (4)$$

2.2. Conventional TSM Controller Design

In this subsection, the conventional NTSM algorithm is utilized as the speed loop controller for PMSM system. Defining the reference mechanical angular speed as Ω_r , then the speed error Ω_e can be written as

$$\Omega_e = \Omega_r - \Omega_n \quad (5)$$

To obtain excellent tracking performance, the terminal sliding surface is given as

$$s = \int \Omega_e dt + \beta [\Omega_e]^\alpha \quad (6)$$

According to [16], the NTSM controller can be designed as

$$T_e^* = \frac{1}{b_0} \left(\dot{\Omega}_r + \frac{1}{\alpha\beta} [\Omega_e]^{2-\alpha} + k \cdot \text{sign}(s) \right) \quad (7)$$

where $\beta > 0$ and $1 < \alpha < 2$.

Assuming that $|d_t| \leq D^+$ and D^+ is a positive constant, then the speed error Ω_e can converge to zero in finite time if the control gain satisfies $k > D^+$ [22].

In order to guarantee the stability of the PMSM drive system, the control gain of conventional NTSM controller should always be larger than the upper bound of total disturbance. In addition, the control performance of conventional NTSM controller with fixed control gain is unsatisfactory in the presence of time-varying disturbance. Thus, it is essential to develop an adaptive law to adjust the control gain of the feedback controller online.

3. Proposed Speed Loop Control Strategy

3.1. Barrier Function Based Adaptive NTSM Controller

By constructing an adaptive law to adjust the control gain of conventional NTSM algorithm, the ANTSM controller is constructed as

$$T_e^* = \frac{1}{b_0} \left(\dot{\Omega}_r + \frac{1}{\alpha\beta} [\Omega_e]^{2-\alpha} + \hat{k}(t) \cdot \text{sign}(s) \right) \quad (8)$$

where $\hat{k}(t)$ is the variable control gain.

The basic strategy of the proposed adaptive algorithm is to first increase the variable control gain $\hat{k}(t)$ until s reaches the region $[-\frac{\tau}{2}, \frac{\tau}{2}]$ at some time $\bar{t}$. Then, the barrier function is adopted to adjust the variable control gain $\hat{k}(t)$ in real time, which ensures that the tracking error remains within a predefined neighborhood of origin $s \in (-\tau, \tau)$. According to the above analysis, the adaptive algorithm is constructed as follows

$$\hat{k}(t) = \begin{cases} \varphi_1 t + \varphi_0, & 0 \leq t < \bar{t} \\ f(s), & t > \bar{t} \end{cases} \quad (9)$$

where φ_0 and φ_1 are positive constants. The barrier function $f(s)$ is designed as

$$f(s) = \frac{\tau \bar{\varphi}}{\tau - |s|} \quad (10)$$

with $\bar{\varphi}$ being a positive constant.

Meanwhile, the relationship between the sliding variable s and the output of barrier function is shown in Figure 1. Obviously, as the sliding variable s increases within $(0, \tau)$, the output of barrier function tends to infinity. As a result, the anti-disturbance property of the proposed ANTSM controller can be enhanced by increasing the control gain. On the contrary, once the sliding variable s decreases, the output of barrier function will decrease accordingly, thus alleviating the chattering phenomenon caused by large control gain.

Increasing the values of parameters φ_0 and φ_1 can reduce the time for sliding variable s to converge from the initial state to the region $[-\frac{\tau}{2}, \frac{\tau}{2}]$. Appropriate parameters φ_0 and φ_1 need to be chosen to avoid overshoot of the control gain $\hat{k}(t)$ with respect to the bound of total disturbance. The parameter $\bar{\varphi}$ is related to the minimum value of the barrier function. Increasing the value of $\bar{\varphi}$ can obviously optimize the disturbance rejection capacity of the

PMSM drive system. However, an excessively large parameter $\bar{\varphi}$ may cause the controller gain calculated by barrier function to be conservative. The parameter τ determines the predefined region of the sliding variable. Decreasing the value of parameter τ will result in better disturbance rejection property and stronger robustness. However, excessively large control gain $\hat{k}(t)$ calculated by barrier function with small parameter τ may lead to undesired chattering phenomenon.

The sliding variable s will be shown to converge and remain in the region $(-\tau, \tau)$ in this subsection. Since the function $\hat{k}(t) = \varphi_1 t + \varphi_0$ is monotonically increasing, the sliding variable will reach the region $[-\frac{\tau}{2}, \frac{\tau}{2}]$ at some time $\bar{t}$ by choosing appropriate parameters φ_0 and φ_1 .

Next, consider the case when $t > \bar{t}$. In this case, the adaptive algorithm is reduced to $\hat{k}(t) = f(s)$. Inspired by [25], the following Lyapunov function is constructed as

$$V_2(s, f(s)) = \frac{1}{2}s^2 + \frac{1}{2}f^2(s) \quad (11)$$

Evaluating the derivative of $V_2(s, f(s))$ yields

$$\begin{aligned} \dot{V}_2(s, f(s)) &= s\dot{s} + f(s)\dot{f}(s) \\ &= s\alpha\beta|\Omega_e|^{\alpha-1}(-d_t - f(s) \cdot \text{sign}(s)) \\ &\quad + \frac{\tau\bar{\varphi}}{(\tau - |s|)^2} \cdot \text{sign}(s)\dot{s}f(s) \\ &\leq \alpha\beta|\Omega_e|^{\alpha-1}(|d_t||s| - f(s)|s|) \\ &\quad + \frac{\tau\bar{\varphi}}{(\tau - |s|)^2} \alpha\beta|\Omega_e|^{\alpha-1}(|d_t| - f(s))f(s) \\ &\leq -\alpha\beta|\Omega_e|^{\alpha-1}(f(s) - D^+)|s| \\ &\quad - \frac{\tau\bar{\varphi}}{(\tau - |s|)^2} \alpha\beta|\Omega_e|^{\alpha-1}(f(s) - D^+)f(s) \end{aligned} \quad (12)$$

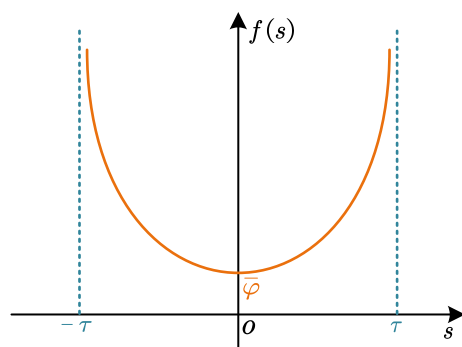


Figure 1. Illustration of barrier function $f(s)$.

To make the expression compact, define

$$\begin{aligned} \eta_1 &= \alpha\beta|\Omega_e|^{\alpha-1}(f(s) - D^+), \\ \eta_2 &= \frac{\tau\bar{\varphi}}{(\tau - |s|)^2} \alpha\beta|\Omega_e|^{\alpha-1}(f(s) - D^+) \end{aligned} \quad (13)$$

Since $D^+ > 0$, an intermediate variable σ is introduced to prove that s will remain in the region $(-\tau, \tau)$ when $t > \bar{t}$. The intermediate variable is designed as

$$\sigma = \begin{cases} \tau \left(1 - \frac{\bar{\varphi}}{D^+}\right), & \text{if } \bar{\varphi} < D^+ \\ 0, & \text{if } \bar{\varphi} \geq D^+ \end{cases} \quad (14)$$

It can be deduced from (10) that $f(s)$ is monotonically increasing on $[0, \tau)$. When $|s| > \sigma$, one has by (14) that $f(s) > f(\sigma) \geq D^+$. Hence, η_1 and η_2 are positive. Under this case, one obtains from (12) and (13) that

$$\begin{aligned} \dot{V}_2(s, f(s)) &\leq -\eta_1 \sqrt{2} \frac{|s|}{\sqrt{2}} - \eta_2 \sqrt{2} \frac{|f(s) - f(0)|}{\sqrt{2}} \\ &\leq -\Pi \left(\frac{|s|}{\sqrt{2}} + \frac{|f(s) - f(0)|}{\sqrt{2}} \right) \\ &\leq -\Pi V_2^{\frac{1}{2}}(s, f(s)) \end{aligned} \quad (15)$$

where $\Pi = \min\{\eta_1 \sqrt{2}, \eta_2 \sqrt{2}\}$.

Obviously, Equation (15) satisfies the finite-time stability criterion in [16]. As a result, the sliding variable s can finite-time converge to the region $[-\sigma, \sigma]$, which is a subset of $(-\tau, \tau)$.

3.2. Design of ARSMO

Due to the fact that the time-varying total disturbance and high-frequency measurement noise always exist in practical engineering, the ARSMO is utilized to estimate the total disturbance and suppress the measurement noise, which performs as the feed-forward compensation in the control design. Assuming that there exists a positive constant d^+ such that $|\dot{d}_t| \leq d^+$, then the ARSMO can be constructed as

$$\begin{cases} \dot{\hat{\Omega}}_n = v_0 \\ v_0 = -\lambda_3 L^{1/3} |\hat{\Omega}_n - \dot{\Omega}_n|^{2/3} \cdot \text{sign}(\hat{\Omega}_n - \dot{\Omega}_n) + \hat{\Omega}_n \\ \dot{\hat{\Omega}}_n = v_1 + b_0 T_e^* \\ v_1 = -\lambda_2 L^{1/2} |\hat{\Omega}_n - v_0|^{1/2} \cdot \text{sign}(\hat{\Omega}_n - v_0) + \hat{d}_t \\ \dot{\hat{d}}_t = -\lambda_1 L \cdot \text{sign}(\hat{d}_t - v_1) \end{cases} \quad (16)$$

where $\hat{\Omega}_n$ is the augmented state of the motor speed Ω_n , $\hat{\Omega}_n$, $\hat{\Omega}_n$ and $\hat{d}_t$ denote the estimation values of $\dot{\Omega}_n$, Ω_n and d_t , respectively. Besides, λ_1 , λ_2 , λ_3 are the positive observer gains and L being a Lipschitz constant for total disturbance d_t . In particular, the positive observer gains can be selected as $\lambda_1 = 1.1$, $\lambda_2 = 3$, and $\lambda_3 = 5$, respectively [26].

Then, we will demonstrate that the actual motor speed and total disturbance can be estimated by ARSMO in finite time. First of all, setting $\phi_0 = \frac{\hat{\Omega}_n - \dot{\Omega}_n}{L}$, $\phi_1 = \frac{\hat{\Omega}_n - \Omega_n}{L}$, and $\phi_2 = \frac{\hat{d}_t - d_t}{L}$, one has

$$\begin{aligned} \dot{\phi}_0 &= \frac{\hat{\Omega}_n - \dot{\Omega}_n}{L} \\ &= \frac{-\lambda_3 L^{1/3} |\hat{\Omega}_n - \dot{\Omega}_n|^{2/3} \cdot \text{sign}(\hat{\Omega}_n - \dot{\Omega}_n) + \hat{\Omega}_n - \dot{\Omega}_n}{L} \\ &= -\lambda_3 \frac{L^{1/3} |\phi_0|^{2/3} \cdot \text{sign}(\phi_0)}{L} + \phi_1 \\ &= -\lambda_3 |\phi_0|^{2/3} \cdot \text{sign}(\phi_0) + \phi_1 \end{aligned} \quad (17)$$

and

$$\begin{aligned}\dot{\phi}_1 &= \frac{\dot{\hat{\Omega}}_n - \dot{\Omega}_n}{L} \\ &= \frac{-\lambda_2 L^{1/2} |\hat{\Omega}_n - v_0|^{1/2} \cdot \text{sign}(\hat{\Omega}_n - v_0) + \hat{d}_t + b_0 T_e^* - \dot{\Omega}_n}{L} \\ &= \frac{-\lambda_2 L^{1/2} |\hat{\Omega}_n - v_0|^{1/2} \cdot \text{sign}(\hat{\Omega}_n - v_0)}{L} + \phi_2\end{aligned}\quad (18)$$

According to (17), it can be deduced that

$$\begin{aligned}\hat{\Omega}_n - v_0 &= \lambda_3 L^{1/3} |\hat{\Omega}_n - \dot{\Omega}_n|^{2/3} \cdot \text{sign}(\hat{\Omega}_n - \dot{\Omega}_n) \\ &= \lambda_3 L^{1/3} |L\phi_0|^{2/3} \cdot \text{sign}(L\phi_0) \\ &= \lambda_3 L |\phi_0|^{2/3} \cdot \text{sign}(\phi_0) \\ &= L(\phi_1 - \dot{\phi}_0)\end{aligned}\quad (19)$$

Together with (18) and (19), it yields

$$\dot{\phi}_1 = -\lambda_2 |\phi_1 - \dot{\phi}_0|^{1/2} \cdot \text{sign}(\phi_1 - \dot{\phi}_0) + \phi_2 \quad (20)$$

Thus, one obtains

$$\begin{aligned}\dot{\phi}_2 &= \frac{\hat{d}_t - d_t}{L} \\ &= \frac{-\lambda_1 L \cdot \text{sign}(\hat{d}_t - v_1) - \hat{d}_t}{L} \\ &= -\lambda_1 \cdot \text{sign}(\hat{d}_t - v_1) - \frac{\hat{d}_t}{L}\end{aligned}\quad (21)$$

With the help of (19), one has

$$\begin{aligned}\hat{d}_t - v_1 &= \lambda_2 L^{1/2} |\hat{\Omega}_n - v_0|^{1/2} \cdot \text{sign}(\hat{\Omega}_n - v_0) \\ &= \lambda_2 L^{1/2} |L(\phi_1 - \dot{\phi}_0)|^{1/2} \cdot \text{sign}(\phi_1 - \dot{\phi}_0) \\ &= L(\phi_2 - \dot{\phi}_1)\end{aligned}\quad (22)$$

Hence, it can be deduced from (21) that

$$\begin{aligned}\dot{\phi}_2 &= -\lambda_1 \cdot \text{sign}(\phi_2 - \dot{\phi}_1) - \frac{\hat{d}_t}{L} \\ &\in -\lambda_1 \cdot \text{sign}(\phi_2 - \dot{\phi}_1) + [-1, 1]\end{aligned}\quad (23)$$

which is understood in the Filippov sense. Combining (17), (20), and (23) together, one has the following differential inclusion

$$\begin{cases} \dot{\phi}_0 = -\lambda_3 |\phi_0|^{2/3} \cdot \text{sign}(\phi_0) + \phi_1 \\ \dot{\phi}_1 = -\lambda_2 |\phi_1 - \dot{\phi}_0|^{1/2} \cdot \text{sign}(\phi_1 - \dot{\phi}_0) + \phi_2 \\ \dot{\phi}_2 \in -\lambda_1 \cdot \text{sign}(\phi_2 - \dot{\phi}_1) + [-1, 1] \end{cases} \quad (24)$$

According to the detailed theoretical analysis in [27], the estimation error can converge to zero in finite time. On this basis, the proposed ANTSM+ARSMO controller can be given as

$$T_e^* = \frac{1}{b_0} \left(\dot{\Omega}_r + \frac{1}{\alpha\beta} [\Omega_e]^{2-\alpha} + \hat{k}(t) \cdot \text{sign}(s) - \hat{d}_t \right) \quad (25)$$

Meanwhile, the block diagram of the ANTSM+ARSMO controller is shown in Figure 2.

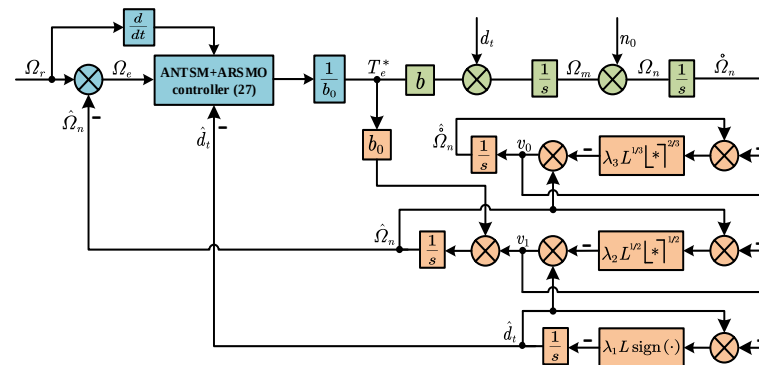


Figure 2. Block diagram of the proposed ANTSM+ARSMO controller.

In order to illustrate the benefits of the proposed ARSMO in suppressing the high-frequency measurement noise, a basic recursive sliding mode observer (RSMO) without the augmented structure is designed to compare with the ARSMO in frequency domain, which is structured as

$$\begin{cases} \dot{\hat{\Omega}}_n = v_0 + b_0 i_q^* \\ v_0 = -\lambda_2 L^{1/2} |\hat{\Omega}_n - \Omega_n|^{1/2} \cdot \text{sign}(\hat{\Omega}_n - \Omega_n) + \hat{d}_t \\ \dot{\hat{d}}_t = -\lambda_1 L \cdot \text{sign}(\hat{d}_t - v_0) \end{cases} \quad (26)$$

Generally, the bode diagram of the observer is employed to analyze its frequency domain characteristics. Since the proposed ARSMO and basic RSMO have nonlinear terms, these transfer functions are difficult to obtain. Consequently, a unified frequency-sweep method is utilized to obtain the bode diagrams of ARSMO and RSMO.

Choosing $L = 200$, the bode diagrams of ARSMO and RSMO are shown in Figure 3. Obviously, the proposed ARSMO performs as a low-pass filter and has satisfactory high-frequency measurement noise suppression property. Let ω_c be the open loop cutoff frequency of the ARSMO, we can find that there is almost no phenomenon of magnitude attenuation and phase lag when $\omega < \omega_c$. Furthermore, the magnitude curve of the ARSMO resembles a slope, and the phase decreases rapidly when $\omega > \omega_c$. Thus, the ARSMO is effective in suppressing high-frequency measurement noise. In addition, the magnitude curve of the ARSMO is below the magnitude curve of the RSMO. Thus, the measurement noise suppression performance of the proposed ARSMO is better than conventional RSMO.

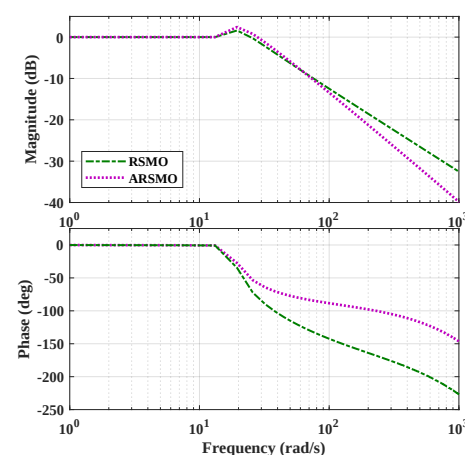


Figure 3. Bode diagrams of ARSMO and RSMO.

3.3. Stability Analysis of Closed-Loop System

Utilizing the estimation values of motor speed and total disturbance, the proposed enhanced ANTSM controller can be given as

$$T_e^* = \frac{1}{b_0} \left(\dot{\Omega}_r + \frac{1}{\alpha\beta} [\hat{\Omega}_e]^{2-\alpha} + \hat{k}(t) \cdot \text{sign}(s) - \hat{d}_t \right) \quad (27)$$

where $\hat{\Omega}_e = \Omega_r - \hat{\Omega}_n$.

The finite time convergence of the ANTSM controller and ARSMO has been demonstrated previously. To prove the stability of the closed-loop system, it is only necessary to prove that the closed-loop system will not diverge to infinity within the time interval $[0, T]$. Next, we will give the rigorous theoretical analysis.

Since $\phi_1 = \frac{\hat{\Omega}_n - \Omega_n}{L}$, we have $\hat{\Omega}_e = \Omega_e - (\hat{\Omega}_n - \Omega_n) = \Omega_e - L\phi_1$. With the help of $\hat{d}_t = d_t + L\phi_2$, the enhanced controller (27) can be rewritten as

$$T_e^* = \frac{1}{b_0} \left(\dot{\Omega}_r + \frac{1}{\alpha\beta} [\Omega_e - L\phi_1]^{2-\alpha} + \hat{k}(t) \cdot \text{sign}(s) - d_t - L\phi_2 \right) \quad (28)$$

By a simple calculation, one has

$$\begin{aligned} \dot{\Omega}_e &= -\frac{1}{\alpha\beta} [\Omega_e - L\phi_1]^{2-\alpha} - \hat{k}(t)\text{sign}(s) \\ &\quad + L\phi_2 - \dot{n}_0 \\ &\leq \frac{1}{\alpha\beta} |\Omega_e - L\phi_1|^{2-\alpha} + |\hat{k}(t)| + L|\phi_2| + |\dot{n}_0| \end{aligned} \quad (29)$$

Choose a Lyapunov function as

$$V_3(\Omega_e) = \frac{1}{2} \Omega_e^2 \quad (30)$$

Differentiating the function $V_3(\Omega_e)$ yields

$$\begin{aligned} \dot{V}_3(\Omega_e) &= \Omega_e \dot{\Omega}_e \text{sign}(\Omega_e) \\ &\leq |\Omega_e| \left(\frac{1}{\alpha\beta} |\Omega_e - L\phi_1|^{2-\alpha} + |\hat{k}(t)| + L|\phi_2| + |\dot{n}_0| \right) \end{aligned} \quad (31)$$

Since $2 - \alpha \in (0, 1)$, one obtains $|\Omega_e - L\phi_1|^{2-\alpha} \leq |\Omega_e|^{2-\alpha} + (L|\phi_1|)^{2-\alpha}$. Due to the fact that ϕ_1 and ϕ_2 can converge to zero in finite time T , ϕ_1 and ϕ_2 are bounded during the time interval $[0, T]$. Meanwhile, the variable control gain $\hat{k}(t)$ and $\dot{n}_0$ are bounded for $t \in [0, T]$. Hence, there exists a positive constant ϵ_1 such that

$$\begin{aligned} \dot{V}_3(\Omega_e) &\leq |\Omega_e| \left(\frac{1}{\alpha\beta} |\Omega_e|^{2-\alpha} + \epsilon_1 \right) \\ &= \sqrt{2} V_3^{\frac{1}{2}}(\Omega_e) \left(\frac{2^{\frac{2-\alpha}{2}}}{\alpha\beta} V_3^{\frac{2-\alpha}{2}}(\Omega_e) + \epsilon_1 \right). \end{aligned} \quad (32)$$

When $V_3(\Omega_e(t)) < 1$, the speed tracking error $\Omega_e(t)$ is obviously bounded. On the contrary, when $V_3(\Omega_e(t)) \geq 1$, one can obtain from (32) that

$$\dot{V}_3(\Omega_e) \leq \sqrt{2} \left(\frac{2^{\frac{2-\alpha}{2}}}{\alpha\beta} + \epsilon_1 \right) V_3^{\frac{3-\alpha}{2}}(\Omega_e) \quad (33)$$

where $\frac{3-\alpha}{2} \in (\frac{1}{2}, 1)$. Then, it can be deduced from (33) that

$$V_3(\Omega_e(t)) \leq \left(\sqrt{2} \left(\frac{2^{\frac{2-\alpha}{2}}}{\alpha\beta} + \epsilon_1 \right) \left(\frac{-1+\alpha}{2} \right) t + V_3^{-\frac{1+\alpha}{2}}(\Omega_e(0)) \right)^{\frac{2}{-1+\alpha}} \quad (34)$$

Consequently, the speed tracking error $\Omega_e(t)$ is bounded and does not diverge within the time interval $[0, T]$.

According to (6), one obtains

$$|s(t)| \leq |\Omega_e(t)|t + \beta|\Omega_e(t)|^\alpha \quad (35)$$

Due to the fact that parameter β is a positive constant and $1 < \alpha < 2$, the sliding variable s is bounded and does not diverge for $t \in [0, T]$.

4. Experiment Results

In order to investigate the advantages of the proposed control scheme, a series of comparative experiments are conducted on the test platform based on dSPACE DS1202 in this subsection. The experimental setup is shown in Figure 4, and the nominal parameters of the test motor are listed in Table 1. The other identical motor is mechanically coupled with the test motor to generate the load torque. The dc-bus voltage is 150 V. The saturation limit of q -axis reference current is 9 A. The field-oriented control method is employed to control the test motor, and the control period is 10 kHz. Decoupled PI regulators are employed in the current-loop to control i_d and i_q . Meanwhile, the well-known ADRC method is employed to compare with the proposed control scheme.

The value of parameter L in RSMO and ARSMO is selected as $L = 200$. The values of parameters α and β in NTSM and ANTSM controllers are chosen as $\alpha = 1.5$ and $\beta = 1$, respectively. The control gain of conventional NTSM controller is selected as $k = 180$, while the control gain of the proposed ANTSM controller is generated by the adaptive algorithm. The parameters of the adaptive law are chosen as $\tau = 3$, $\varphi_0 = 50$ and $\varphi_1 = 20$.

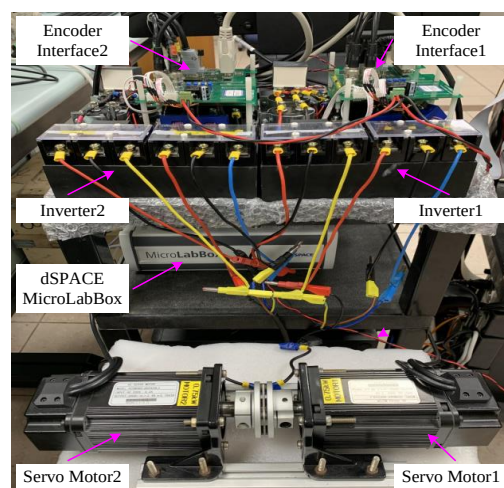


Figure 4. PMSM experimental bench.

Table 1. Main parameters of the pmsm system.

Symbol	Parameter	Value	Unit
R_s	Stator resistance	1.1	Ω
L_s	dq -axis inductance	5.7	mH
φ_f	Rotor flux linkage	0.092	wb
P_N	Rated power	0.75	kW
T_N	Rated torque	2.4	N·m
J	Moment of inertia	1.62×10^{-4}	kg/m ²
U_n	DC-bus rated voltage	220	V
p_n	Number of pole pairs	4	-

To illustrate the superior performance of the proposed ANTSM+ARSMO control scheme, experiments are implemented among the ADRC controller, NTSM+RSMO controller, ANTSM+RSMO controller, and the proposed ANTSM+ARSMO controller. The corresponding experimental results are given in Figures 5 and 6. Figure 5 shows the disturbance rejection properties of these four control algorithms. The experimental results from top to bottom include the reference motor speed, actual motor speed, q -axis current, and A -phase current, respectively. It can be seen from Figure 5a,b that the conventional ADRC method with linear terms and conventional NTSM controller with fixed control gain is insufficient to suppress the adverse effect of load torque variation. Due to the fact that the NTSM+RSMO controller adopts a sufficiently large control gain to suppress the time-varying total disturbance, the chattering phenomenon during the stable operation of the motor is larger than that under the ANTSM controller. According to the experimental results of q -axis current, it can be concluded that the proposed ARSMO has better high-frequency measurement noise suppression performance compared to the conventional RSMO. The comparative results of the four controllers under load variation are summarized in Table 2.

Table 2. Comparison of performance indices under load variation.

Controller	Speed Drop/(rpm)	Current Ripple/(A)	Settling Time/(s)
ADRC	62	0.35	0.14
NTSM+RSMO	55	0.6	0.11
ANTSM+RSMO	28	0.48	0.12
ANTSM+ARSMO	24	0.42	0.12

Figure 6 demonstrates the tracking performance of these four control strategies. The experimental results from top to bottom include the reference motor speed, actual motor speed, speed tracking error, and q -axis current, respectively. The reference motor speed signal is chosen as $\Omega_r = 800 \sin(10\pi t)$. Owing to the utilization of barrier function to adjust the control gain online, the speed tracking error under the proposed ANTSM controller is significantly smaller than that of the conventional NTSM controller. In addition, the maximum speed tracking error under the proposed ANTSM+ARSMO controller is almost the same as that under ANTSM+RSMO controller. However, it can be clearly observed from the experimental results of q -axis current, the control performance of the ANTSM+ARSMO control strategy is better than that of ANTSM+RSMO control algorithm. The comparative results of the four controllers under sinusoidal reference are summarized in Table 3.

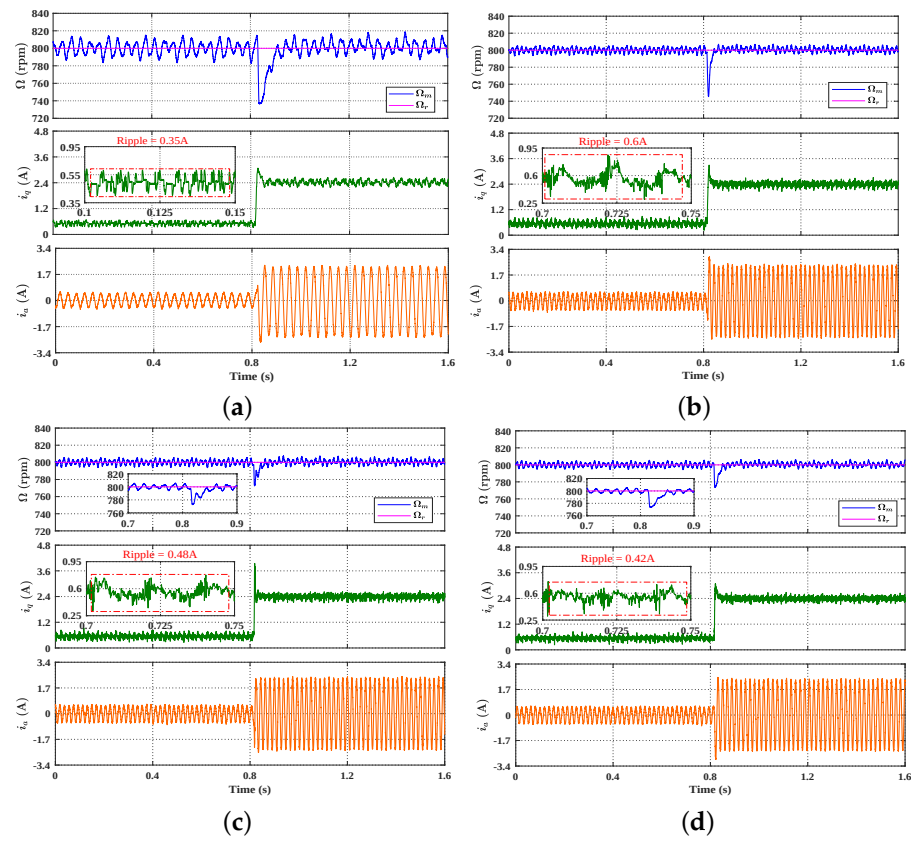


Figure 5. Disturbance rejection performance of four control methods. (a) ADRC. (b) NTSM+RSMO. (c) ANTSM+RSMO. (d) ANTSM+ARSMO.

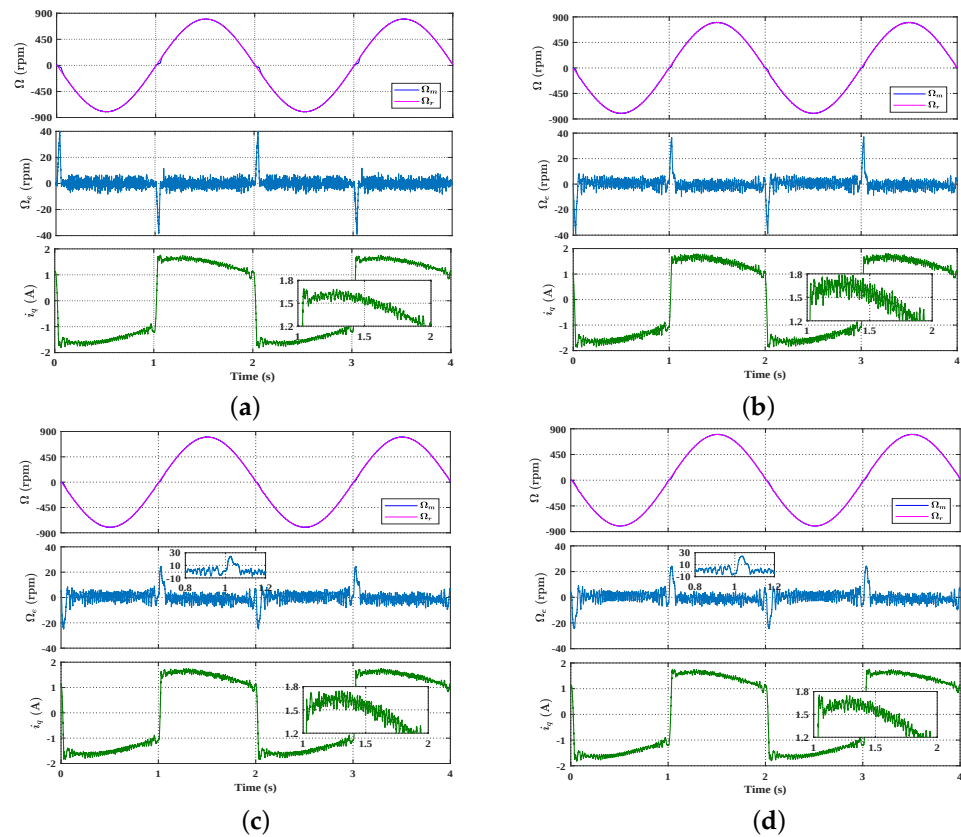
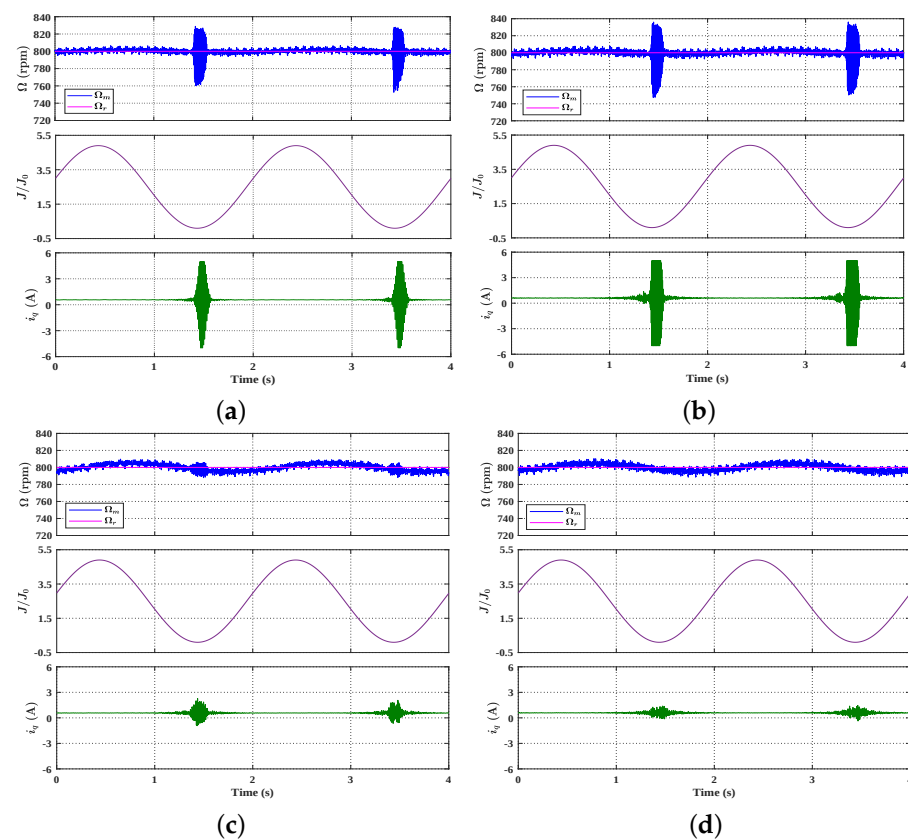


Figure 6. Tracking performance of four control methods under sinusoidal reference. (a) ADRC. (b) NTSM+RSMO. (c) ANTSM+RSMO. (d) ANTSM+ARSMO.

Table 3. Comparison of performance indices under sinusoidal reference.

Controller	Maximum Tracking Error/(rpm)	Current Ripple/(A)
ADRC	62	Small
NTSM+RSMO	55	Large
ANTSM+RSMO	28	Medium-large
ANTSM+ARSMO	24	Medium-small

Figure 7 shows the robustness of these four control methods. The experimental results from top to bottom include the reference motor speed, actual motor speed, ratio of actual inertia to nominal inertia, and q -axis current, respectively. Obviously, the conventional NTSM+RSMO control method has the worst robustness. The motor has severe oscillation phenomenon during operation, and the actual amplitude of the q -axis reaches its limit. At the same time, the actual motor speed under the ANTSM+RSMO controller also has slight oscillation phenomenon. Overall, the proposed ANTSM+ARSMO control strategy has satisfactory robustness in the presence of inertia mismatch.


Figure 7. Robustness of four control methods. (a) ADRC. (b) NTSM+RSMO. (c) ANTSM+RSMO. (d) ANTSM+ARSMO.

The experimental results of ANTSM+ARSMO control method under different values of parameter τ are shown in Figures 8 and 9. The values of parameter τ are chosen as 2.5, 3.5, and 4.5, respectively. According to Figure 8, it is clear that decreasing parameter τ can diminish the speed drop under load torque variation. Since the value of parameter τ is related to the range of speed tracking error, a smaller parameter τ can improve the disturbance rejection performance of the PMSM system. Similarly, as shown in Figure 9, decreasing the value of parameter τ can reduce the motor speed tracking error under the sinusoidal reference speed signal. However, the change of parameter τ has little effect on

the current ripple. Overall, the variation of parameter τ causes a important impact on the dynamic performance of the PMSM drive system. The comparative results of the proposed controller under different parameter τ are summarized in Table 4.

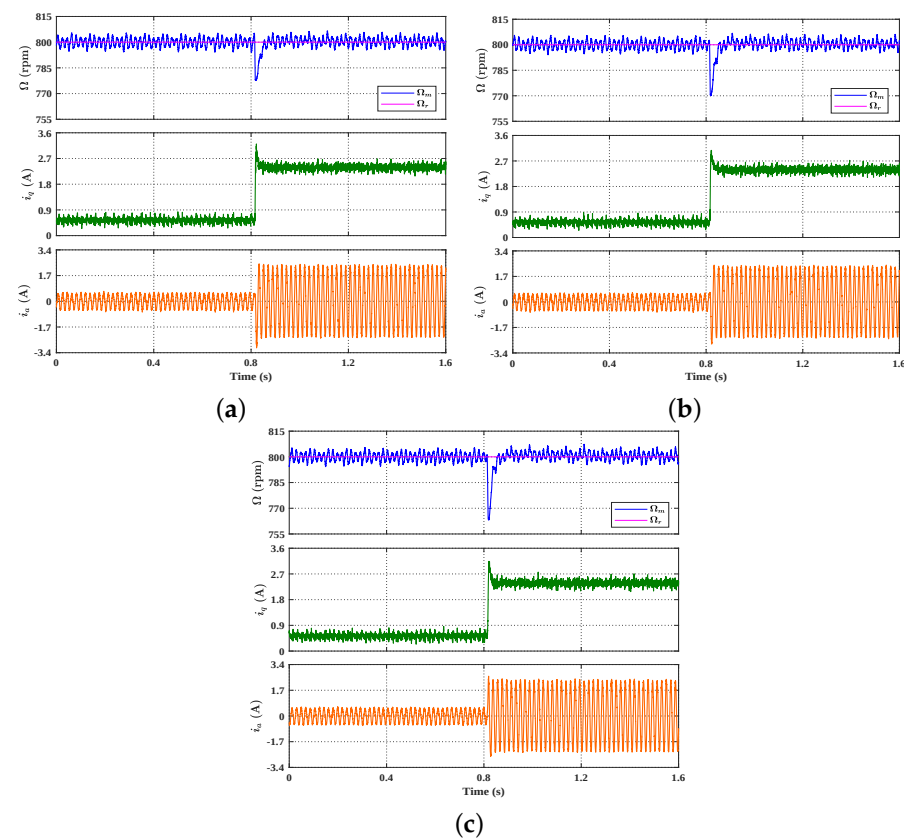


Figure 8. Disturbance rejection performance of ANTSM controller under different τ . (a) $\tau = 2.5$. (b) $\tau = 3.5$. (c) $\tau = 4.5$.

Table 4. Comparison of performance indices under different parameter τ .

Value	Speed Drop/(rpm)	Settling Time/(s)	Maximum Tracking Error/(rpm)
$\tau = 2.5$	22	0.09	21
$\tau = 3.5$	30	0.11	27
$\tau = 4.5$	36	0.12	35

Figures 10 and 11 illustrate the performance of ANTSM+ARSMO control algorithm under different parameter $\bar{\varphi}$. The values of $\bar{\varphi}$ are selected as 80, 160, and 240, respectively. According to the experimental results of actual motor speed in Figure 10 and speed tracking error in Figure 11, it is evident that increasing parameter $\bar{\varphi}$ can diminish the speed drop under load torque variation and the speed tracking error under sinusoidal reference signal at the same time. Since the value of parameter $\bar{\varphi}$ is related to the minimum value of control gain, increasing parameter $\bar{\varphi}$ may cause the chattering phenomenon, which is reflected in the experimental results of the q -axis current ripple. The comparative results of the proposed controller under different parameter $\bar{\varphi}$ are summarized in Table 5.

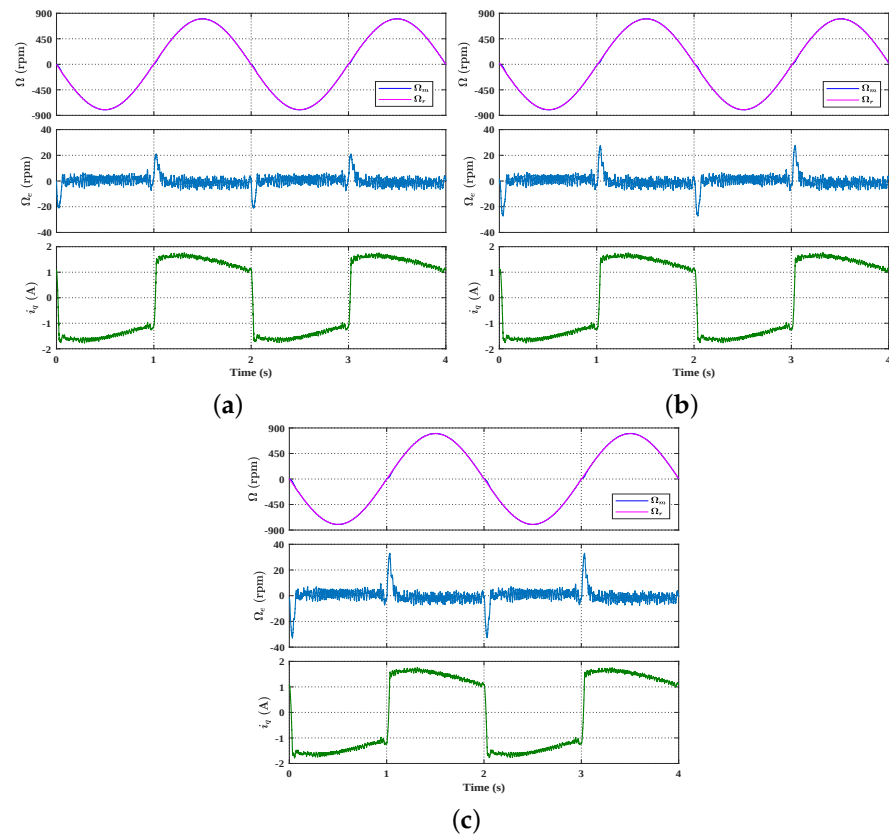


Figure 9. Tracking performance of ANTSM controller under different τ . (a) $\tau = 2.5$. (b) $\tau = 3.5$. (c) $\tau = 4.5$.

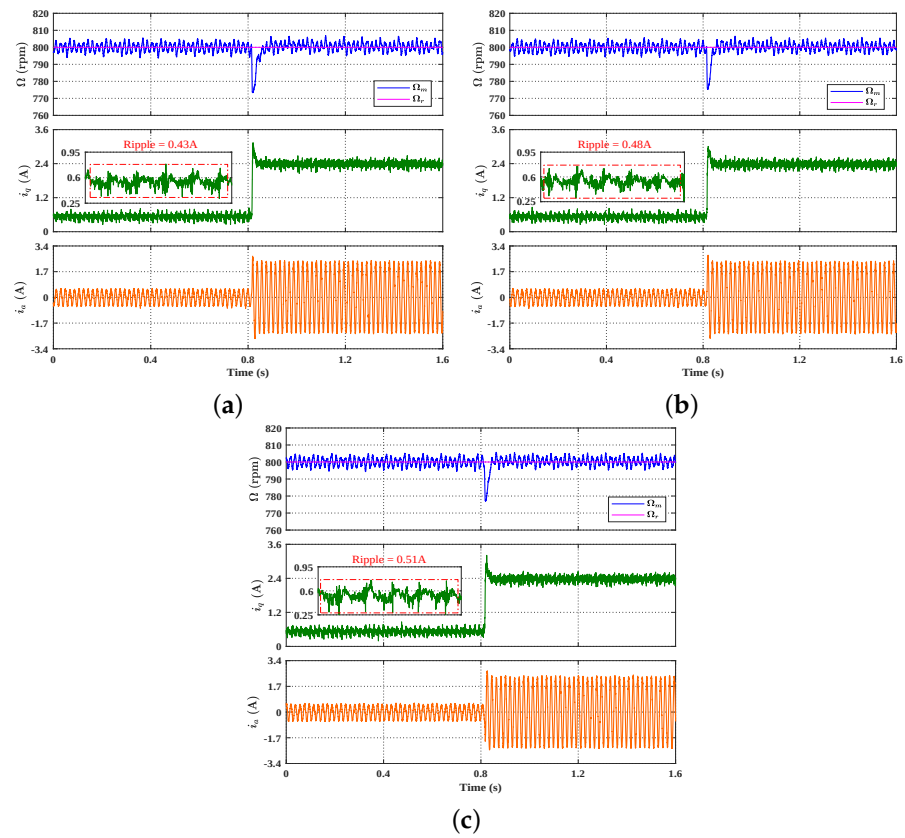


Figure 10. Disturbance rejection performance of ANTSM controller under different $\bar{\varphi}$. (a) $\bar{\varphi} = 80$. (b) $\bar{\varphi} = 160$. (c) $\bar{\varphi} = 240$.

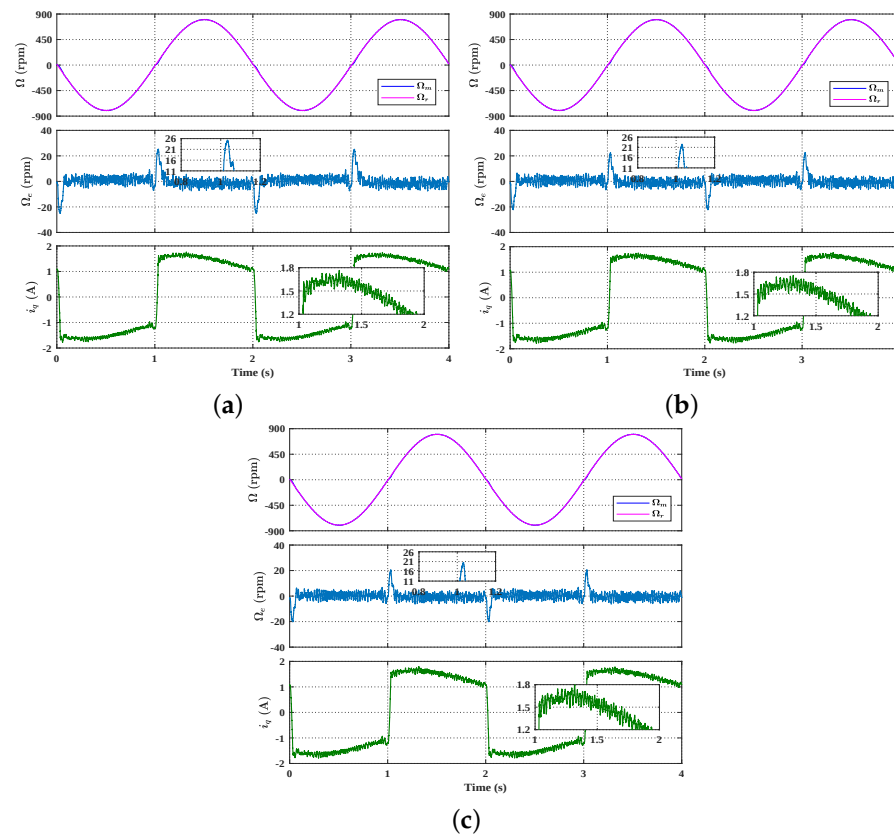


Figure 11. Tracking performance of ANTSM controller under different $\bar{\varphi}$. (a) $\bar{\varphi} = 80$. (b) $\bar{\varphi} = 160$. (c) $\bar{\varphi} = 240$.

Table 5. Comparison of performance indices under different parameter $\bar{\varphi}$.

Value	Speed Drop/(rpm)	Current Ripple/(A)	Maximum Tracking Error/(rpm)
$\bar{\varphi} = 80$	26	0.43	25
$\bar{\varphi} = 160$	23	0.48	23
$\bar{\varphi} = 240$	22	0.51	20

5. Conclusions

This paper presents a combination of the ANTSM controller and ARSMO algorithm for PMSM speed regulation system. By employing the barrier function to adjust the control gain of conventional NTSM controller, the dynamic performance of PMSM system can be significantly improved without sacrificing its steady-state performance. Meanwhile, the adverse effect of measurement noise generated by position sensors is alleviated by adopting the ARSMO with filtering characteristics, while avoiding excessive feedback control gain. The comprehensive experimental results illustrate that the proposed control strategy can effectively suppress the total disturbance and measurement noise compared with conventional NTSM controller and basic RSMO without the augmented structure. Our future work will focus on verifying the effectiveness of the proposed algorithm in more complex industrial environments and at low sampling frequencies.

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